

Task – 1

Titanic Survival Analysis – Report

1. Data Loading and Overview

The Titanic dataset was loaded using pandas: python

```
df = pd.read_csv('train.csv')
```

Basic inspection was done using: python

```
df.head()
```

```
df.info()
```

```
df['Name'].unique()
```

This helped understand the dataset structure and identify missing values.

2. Missing Value Handling

Missing values were addressed as follows:

- **Age:** Filled with the **median** age: python

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

- **Cabin:** Dropped completely due to excessive missing data: python

```
df.drop(columns=['Cabin'], inplace=True)
```

- **Embarked:** Filled with the **most common value** (mode): python

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

3. Visualization

A basic line plot was used to visualize the **Age** distribution: python

```
plt.plot(df['Age'])
```

```
plt.xlabel('PERSONS')
```

```
plt.ylabel('AGES')
```

4. Summary

- The dataset was cleaned by handling missing values.
- Non-useful or sparse columns like Cabin were removed.
- A simple visualization of age was generated.
- Dataset is now preprocessed and ready for further analysis or modeling.